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# The barbell and the dark horse

A thought exercise on convex tail sleeves — Taleb's barbell, its reversed *curveball*, and an honest measurement of what convexity costs and when it pays. This is research, not a deployed strategy.

by **Carter Warrens** · research track · 2026 · companion to “Harvesting Risk, Not Return”

**RESEARCH NOTE — NOT DEPLOYED**

Nothing here manages capital or is a recommendation. Figures use real instruments as honest proxies — **BIL** (T-bills) for the safe sleeve, **VIXY** (short-term VIX futures) for the convex sleeve, 2016–2026. Read the *shapes* (skew, worst-month, crisis capture), not the point estimates: a tail strategy's backtest is dominated by whether a large tail happened to fall inside the window — and the real tail is always worse than the sample shows.

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ABSTRACT

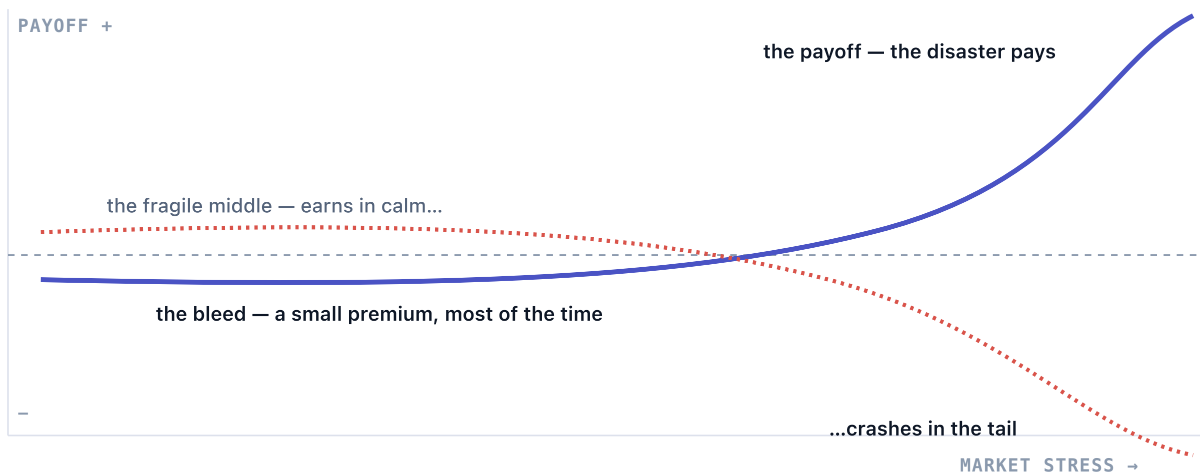
The barbell avoids the fragile middle: hold most of the book in something that cannot be wiped out, and a little in something maximally *convex* — capped downside, uncapped upside. We measure the classic 90/10, a growth-plus-hedge variant, and a reversal we call the **curveball** (90% in the convex sleeve). The result is a clean map: convexity is either **free** — funded by the momentum premium, via trend — or **paid** — funded by decay, via long-vol. The curveball is a dark horse: you let it bleed slowly and hold it for the disaster, where it pays for everything. Every attempt to time or trim the bleed also removes the payoff, because they are the same thing.

## 1 Convexity, not volatility

The barbell's power is not "90% boring, 10% exciting." It is a specific payoff shape. The tail sleeve must be *convex* — you can lose only the premium, but win many multiples — and the safe sleeve must be genuinely safe, because staying solvent is what lets convexity pay. Decomposed:

$$r^{\text{barbell}} = \underbrace{w_s\, r_{\text{bills}}}_{\text{no ruin}} + \underbrace{w_c\, r_{\text{convex}}}_{\text{bleeds } -b \text{ most of the time, pays } +g \text{ in the disaster}}$$

FIGURE — THE TWO PAYOFF SHAPES




The convex sleeve (solid) and the fragile "middle" like plain equities (dashed). The barbell deliberately owns the first shape and avoids the second. **The whole game is the shape of the tail, not the size of the wiggle.**

## 2 What it costs, and when it pays

Measured with real proxies (BIL + VIXY, 2016–2026). Barbells optimize convexity and anti-ruin, *not* Sharpe, so the honest columns are skew, worst month, and crisis capture:

| PORTFOLIO | CAGR | MAX DD | SKEW (MO) | WORST MO | COVID CRASH |
|---|---|---|---|---|---|
| Safe — 100% bills | +2.1% | 0% | +0.35 | 0% | 0% |
| Fragile middle — 100% equity | +15.0% | −34% | −0.41 | −13% | −33% |
| **Classic barbell — 90/10** | −3.0% | −32% | **+2.09** | **−4%** | **+19%** |
| Growth + hedge — 90% eq / 10% vol | +8.7% | −20% | −0.03 | −7% | −15% |
| **Curveball — reversed 10/90** | **−44%** | **−100%** | **+2.16** | −32% | **+237%** |

The classic barbell *bled* −3%/yr across a mostly-calm decade — but its shape is unmistakable: skew +2.1, worst month just −4%, and it made money in *every* crisis (+19% in the COVID crash while equities lost −33%). The negative carry is the premium; the convexity is real. The growth-plus-hedge row shows the practical use — a 10% vol sleeve on an equity book cut the drawdown −34% → −20% and flipped the skew — the barbell's value is what it does to a *portfolio's* left tail, not its standalone return.

## 3 The dark horse

The curveball reverses the weights — 90% in the convex sleeve. Held and compounded it is **ruin**: −44%/yr, −100% drawdown (you cannot hold 90% of a product that decays ~99% over a decade). But that framing misses the point. The curveball is not a book you compound; it is the small sliver you *let bleed* and keep for the disaster. Its convex sleeve returned **+237% in the COVID crash** — on a 10% weight, **+23% to a whole portfolio**, enough to cover a very bad year. That is the entire job.

$$\mathbb{E}[r_{\text{convex}}] = p\,g - (1-p)\,b, \qquad \text{positive skew when } g \gg b,\ p \text{ small}$$

And crucially — **you cannot improve it, and you shouldn't try.** We tested the obvious rules. Deploying it only when volatility is cheap halved the bleed (−44% → −19%) but *removed the crash payoff* (COVID +237% → 0%; the skew went negative) — the gate exits as vol rises, i.e. right before the payoff. Selling into spikes cut COVID from +237% to +17%. Every rule that trims the bleed also kills the tail, because **the bleed and the convexity are the same thing** — both come from being maximally positioned in a decaying convex instrument at all times. Timing the tail removes the tail. You hold it, you let it bleed, and you let the disaster pay. That is the discipline, and it is the whole strategy.

## 4 Free convexity vs paid convexity

Stepping back, convexity comes in exactly two forms — and this is the note's one durable conclusion:

| | FREE CONVEXITY | PAID CONVEXITY |
|---|---|---|
| Instrument | trend / time-series momentum | long volatility (options, VIX) |
| Funded by | the momentum premium | option / futures decay |
| Protects against | **sustained** crises (e.g. 2022) | **sharp** crashes (e.g. COVID) |
| Carry cost | ~none (it *improves* return) | ruinous bleed (≈ −44%/yr) |
| Can you time it? | it *is* the signal | no — timing removes the payoff |

The two do different jobs. Trend-following is a convex, momentum-*funded* hedge that pays in drawn-out crises — but it is structurally too slow for a four-week crash-and-recovery, at any speed we tested. Long-vol is the only thing that pays on the crash *day* — but it bleeds, and it cannot be cheapened by timing. There is no configuration in this space that delivers cheap, well-timed, sharp-crash protection. That is not a search we gave up on; it is the shape of the problem.

## 5 The honest use

For a systematic book like the Spine, the conclusion is practical. The Spine already owns the *free* convexity — its trend sleeve is a crisis hedge funded by momentum, which is exactly why bolting a paid long-vol overlay on top only deepened its drawdown. Sustained-crisis protection is already in the price. **Sharp-crash protection is the one thing money can't buy cheaply** — so if you want it, buy it as a small, budgeted, always-bleeding sliver you have decided in advance you can lose, sized as insurance and never optimized. That is the dark horse, held for the day it earns its keep. The mistake is not paying the premium; the mistake is trying to be clever about when.

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Reproducible sketches: **scripts/research/barbell_sketch.py**, **spine_tail_overlay_sketch.py**, **convex_trend_sketch.py**, **crash_and_curveball_sketch.py**. Validated strategy: **docs/whitepaper.html**. Research architecture: **docs/CANONICAL_ARCHITECTURE.md**.